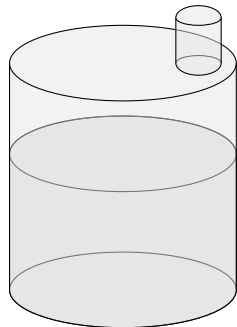


6.4 Work - continued

Pumping liquids

When calculating work done by pumping liquids we will approximate the work by dividing the tank into n layers.



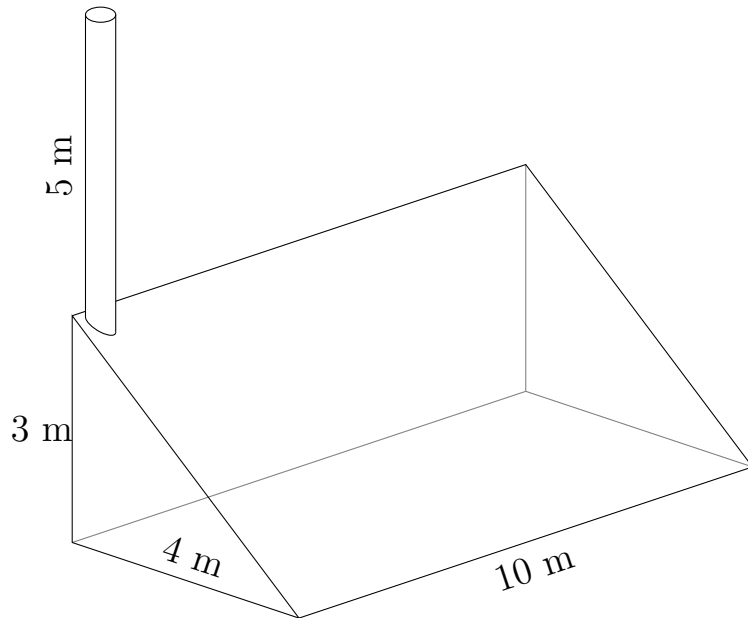
- How can we find the mass of each of the n layers?
- Given mass, $m(x_i^*)$, how can we find the force?
- How can we find the work to pump out the liquid?

Example 1

A rectangular tank full of water has height 12 ft, width 3 ft, and length 5 ft. Find the work done in pumping all of the water to the top of the tank given that the density of water is 62.4 lbs/ft^3 .

Example 2

A tank pictured below is full of water. The density of water is 1000 kg/m^3 . Set up the integral to find the work needed to pump the water to a height of 5 m above the top of the tank.



Summary for Pumping Liquids

1. How do I know what technique to use to find the cross-sectional area?
2. What do the bounds of the integral represent?
3. What is the distance that a cross-section of liquid needs to travel?

Extra Example

A spherical tank with radius 10 ft is half full of water. Set up an integral to find the work needed to pump all of the water out of a spout, which is length 3 ft, at the top of the sphere.